

JPE Replication Report

PUBLISHED
July 2, 2026

1 Paper Identification

ID	20231305
Author	Sullivan
Title	Fee Optimality in a Two-Sided Market
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. We will attempt a remote replication as mentioned via email.
2. There are a few potential items to fix below, please consider those, resubmit, and we take it from there.

3 General

- ☑ Data Availability and Provenance Statements
 - ☑ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☑ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☑ License for Data (optional, but recommended)
 - ☑ Details on each Data Source
- ☑ Dataset list
- ☑ Computational requirements
 - ☑ Software Requirements
 - ☑ Controlled Randomness (as necessary)
 - ☑ Memory, Runtime, and Storage Requirements
- ☑ Description of programs/code
 - ☑ License for Code (Optional, but recommended)
- ☑ {REPLICATOR} to Replicators
 - ☑ Details (as necessary)
- ☑ List of tables and programs
- ☑ References (Optional, but recommended)

Comment: My compliments on the README, it is extremely detailed and easy to understand!

4 High-level Description of Research Project

This research project...

- ☑ uses observational data.
- ☑ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).

- ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
- ☐ A PDF copy of IRB approval is contained in the package.
- ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
- ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. - As per the [JPE policy](#), "In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

{REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
ACS aggregate Subject Tables (5-year, 2015-2019)	yes	no	yes	yes	yes	yes
ACS microdata via IPUMS USA (1-year, 2021)	yes	no	yes	yes	yes	yes
OxCGRt COVID-19 policy data	yes	no	yes	yes	yes	yes
Johns Hopkins COVID-19 case data	yes	no	yes	yes	yes	yes
MIT Election Lab county presidential returns	yes	no	yes	yes	yes	yes
SimpleMaps US Zip Code Database (Basic, v1.77)	no	no	yes	yes	yes	yes
Fee-cap legislation panel (hand-collected)	yes	no	no	yes	yes	yes

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
Restaurant price sample (hand-collected)	yes	no	no	yes	yes	yes
Second Measure market shares (hand-transcribed)	no	no	no	yes	yes	yes
Author-constructed lookup tables	yes	no	no	yes	yes	no
Numerator consumer panel	no	no	no	yes	yes	yes
YipitData restaurant listings	no	no	no	yes	yes	yes
Edison transactional panel	no	no	no	yes	yes	yes
InfoGroup / Data Axle restaurant census	no	no	no	yes	yes	yes
Web-harvested platform data (raw layers)	yes	no	no	yes	yes	yes
Web-harvested analysis intermediates (analysis_data/)	no	no	no	yes	yes	yes

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - **R** (the published results were last produced with R 4.5.2)
 - CRAN packages: `AER`, `data.table`, `doBy`, `dplyr`, `fixest`, `geosphere`, `Matrix`, `parallel`, `pracma`, `R.matlab`, `RColorBrewer`, `scales`, `stringdist`, `stringi`, `stringr`, `tidyr`, `wesanderson`, and `noncensus` (a dependency of `EconTools`; archived from CRAN — if `install.packages("noncensus")` fails, `code/0_setup.R` falls back to the version-pinned CRAN archive tarball (`noncensus_0.1.tar.gz`), and failing that to `github.com/ramhiser/noncensus`).
 - **Two locally-developed R packages** at `code/EconTools/` (general file-I/O and formatter helpers) and `code/FoodDeliveryTools/` (analysis-specific helpers: data loaders, equilibrium computations, post-estimation utilities). `code/0_setup.R` installs both from source.
 - **MATLAB** (last run with R2026a)
 - **Parallel Computing Toolbox** (used by the `parfor` loop in `fe_estimation_boot.m`).

- **Statistics and Machine Learning Toolbox** (used by various distribution functions in the simulation routines).
 - **Optimization Toolbox** (referenced by a few auxiliary helpers, though the production GMM estimator does not depend on it).
 - **Stata** (last run with **Stata 19**)
 - `reghdfe` (for the consumer-DiD regressions in `code/explore_yipit/` and the price-effect regressions in `code/numerator_menu_pricing/estimate_price_effects_disagg.do`)
 - `estout` (for table export from Stata)
 - `ftools` (a `reghdfe` dependency)
- ✓ Computational Requirements specified as follows:
- 16-core & 64 GB RAM
- ✓ Time Requirements specified as follows:
- 160 hours
- ✓ Requirements are complete.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 60000
- Variables flagged in data: 346913
- Code files with PII references: 212
- PII references in code: 5174

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	<code>ACSDT5Y2019.B01003_data_with_overlays_2021-06-12T231340.csv</code>	1	name
Data	<code>ACSST5Y2019.S0101_data_with_overlays_2021-11-06T184610.csv</code>	1	name
Data	<code>ACSST5Y2019.S1201_data_with_overlays_2021-11-14T184552.csv</code>	1	name
Data	<code>ACSST5Y2019.S1501_data_with_overlays_2021-11-08T185630.csv</code>	1	name
Data	<code>ACSST5Y2019.S1901_data_with_overlays_2021-11-04T111610.csv</code>	1	name
Data	<code>ACS_one_year_2021_v3.csv</code>	1	child
Data	<code>MC_results.csv</code>	1	lat
Data	<code>0xCGRt_US_latest.csv</code>	5	country, name, school, lat
Data	<code>ZCTAs.csv</code>	2	lat, lon
Data	<code>all_data.csv</code>	7	name, lat, loc, lon
Data	<code>all_data.csv</code>	7	name, lat, loc, lon
Data	<code>all_data.csv</code>	7	name, lat, loc, lon
Data	<code>all_data.csv</code>	7	name, lat, loc, lon

File Type	File	Variables/References	PII Categories
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-06-09T18:09:20.621

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (<code>C:\</code> etc)
238	8	78	9	0

Note: This list contains 620878 entries (109.5 MB) — too large to embed in the report. The full list is available as a GitHub Actions artifact for this run: <https://github.com/JPE-Reproducibility/JPE-Sullivan-20231305/actions/runs/27214314645>

7.1.3 Zero Size Files Report

We found the following zero size files:

Filename	Size (MB)	Checksum (MD5)
/fee_optimality_replication/code/FoodDeliveryTools/R/f1_draw_probabilities_SCF_2016.R	0.0	da39a3ee5e6b4b0d3255bfef95601890afd80709
/fee_optimality_replication/data/web_harvesting/grubhub_CBSAs/muni_details/Astoria,-NY_2021-06-02-19-49.pickle	0.0	da39a3ee5e6b4b0d3255bfef95601890afd80709
/fee_optimality_replication/data/web_harvesting/grubhub_ca-los_angeles/proc_links-ca-los_angeles_v2021-05-31-22-39.pickle	0.0	da39a3ee5e6b4b0d3255bfef95601890afd80709

7.1.4 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/fee_optimality_replication/data/ACS/ACSST5Y2019.S1501_2021-12-27T205749/ACSST5Y2019.S1501_data_with_overlays_2021-11-08T185630.csv	141.94
/fee_optimality_replication/data/web_harvesting/grubhub_CBSAs/processed_details.csv	142.96
/fee_optimality_replication/data/web_harvesting/grubhub_CBSAs/geocoded_details.csv	174.71
/fee_optimality_replication/data/ACS/ACS_one_year_2021_v3.csv	370.06

7.2 Code description

- There is one run_all.sh master script, that calls for: 72 R scripts, 4 Stata do-files, 1 Matlab scripts and 2 Python scripts.

cloc

github.com/AIDanial/cloc v 2.06 T=2196.37 s (27.2 files/s, 6107.8 lines/s)

Language	files	blank	comment	code
CSV	59521	11	0	13382982
R	198	4116	3508	18320
MATLAB	30	580	490	2354
Markdown	8	235	0	1225
Bourne Shell	1	61	205	207
Stata	5	53	12	179
Python	2	43	19	105
Mathematica	1	22	0	83
Text	8	41	0	58
---	---	---	---	---
SUM:	59774	5162	4234	13405513

- ☒
- The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

7.3 Computing Environment of the Replicator

- Nuvolos Cloud 16 vCPU & 64GB RAM
- Stata/MP 19.5
- Matlab R2025b
- R 4.5.2
- VSCode 1.23.0 with Py3.13

8 Replication

- Downloaded code and data via dropbox & git
- Loaded the replication package in VSCode
- Followed the steps described in the README
- Was not able to access all of the the proprietary datasets, therefore the synthetic data generator (generate_synthetic_data.R) was used to simulate the necessary datasets. Ran this code succesfully.
- Ran code/0_setup.R
- Ran run_all.sh. However code broke down in multiple occassions, see explanation below.

9 Findings

Issue 1 - During the first run, the pipeline broke down at `code/compute_pop_distribution/pop_dist_take2.R`, as the script referenced files in the `est_data` directory that had not yet been created at that point in the pipeline. I moved this script below the `prepare_est_data`. However, similar issues occurred with other scripts that referenced to datasets not yet available. To resolve this, the order of scripts in Stage 1 of `run_all.sh` was adjusted as shown below:

```
if stage_should_run 1; then
  banner 1 "Raw-data processing"
  run_r "code/process_acs/process_acs.R"
  run_r "code/process_acs/nearby_demos_acs.R"
  run_r "code/generate_geo/generate_geo.R"
  run_r "code/generate_geo/load_zcta_data.R"
  run_r "code/process_infogroup/process_infogroup.R"
  run_r "code/process_yipit/add_brands.R"
  run_r "code/process_yipit/add_infogroup.R"
  run_r "code/process_yipit/add_store_locations_w_infogroup.R"
  run_r "code/process_yipit/separate_data.R"
  run_r "code/process_yipit/compute_subset.R"
  run_r "code/process_yipit/determine_top_chains_and_cuisines.R"
  run_r "code/process_yipit/produce_DiD_data_v2.R"
  run_r "code/process_numerator/collapse_static.R"
  run "python3 code/process_numerator/identify_static_trans.py"
  run "python3 code/process_numerator/identify_static_trans-summary.py"
  run_r "code/process_numerator/combine_fact_summary.R"
  run_r "code/process_numerator/combine_item_table.R"
  run_r "code/process_numerator/add_category.R"
  run_r "code/process_numerator/static_event_dataset.R"
  run_r "code/process_numerator/produce_nonstatic_table.R"
  run_r "code/process_numerator/prepare_regression_data.R"
  run_r "code/describe_fee_caps/monthly_fee_cap_data.R"
  run_r "code/describe_fee_caps/evolution_of_fee_caps_v2.R"
  run_r "code/restaurant_choice_demos/nearby_demographics.R"
  run_r "code/prepare_est_data/zcta_mapping.R"
  run_r "code/prepare_est_data/zcta_resto_data.R"
  run_r "code/prepare_est_data/produce_month_zip_level.R"
  run_r "code/prepare_est_data/combine_data.R"
  run_r "code/prepare_est_data/check_contradiction.R"
  run_r "code/prepare_est_data/combine_months.R"
  run_r "code/compute_pop_distribution/pop_dist_take2.R"
  run_r "code/zip_code_weights/construct_zip_code_weights.R"
  run_r "code/numerator_menu_pricing/prepare_menu_data.R"
  run_r "code/numerator_menu_pricing/prepare_geo_data.R"
  run_r "code/numerator_menu_pricing/prepare_item_data.R"
  run_r "code/numerator_menu_pricing/merge_item_level_data.R"
  run_manual "Run in Stata: code/numerator_menu_pricing/convert_to_dta.do"
  run_manual "Run in Stata: code/numerator_menu_pricing/estimate_price_effects_disagg.do"
  run_r "code/numerator_menu_pricing/generate_item_level_reg_table.R"
  run_r "code/numerator_menu_pricing/compute_price_indices.R"
fi
```

Issue 2 - Encountered an error in `code/process_yipit/add_brands.R`. `generate_synthetic_data.R` correctly creates separate .rds files per month/year combination and listings_v2.csv, however the generated .rds files contain 0 rows. As a result, `add_brands.R` fails because it cannot process empty files:

```
Error in `$<-data.frame`(`*tmp`, "brand", value = NA) :
  replacement has 1 row, data has 0
Calls: main -> add_brands -> $<- -> $<-data.frame
Execution halted
```

Issue 3 - Encountered an error in `code/process_numerator/combine_item_table.R`. The script hardcodes a path to a proprietary Numerator file that is apparently not generated by `generate_synthetic_data.R`: `inpath.b <- data/numerator/2022-03-04/standard_nmr_feed_item_table.csv`.

```
Error in data.table::fread(inpath.b, sep = "|") :
  File 'data/numerator/2022-03-04/standard_nmr_feed_item_table.csv'
  does not exist or is non-readable.
Execution halted
```

As a result of issue 2 & 3, I was not able to run the first stage of the master script. At least two scripts fail when running the pipeline with synthetic data, making it impossible to fully replicate the results without access to the proprietary data. The synthetic data script was run successfully, but it

seems that the generated output is not sufficient.

9.1 Missing Requirements

- ☐ Software Requirements
 - ☐ Stata
 - ☐ Version
 - Packages go here
 - ☐ Matlab
 - ☐ Version
 - ☐ R
 - ☐ Version
 - R packages go here
 - ☐ Python
 - ☐ Version
 - Python package go here
 - ☐ REPLACE ME WITH OTHER
- ☐ Computational Requirements specified as follows:
 - Cluster size, disk size, memory size, etc.
- ☐ Time Requirements
 - Length of necessary computation (hours, weeks, etc.)

[REQUIRED] Please amend README to contain complete requirements.

You can copy the section above, amended if necessary.

9.2 Data Preparation Code

Program code/0_setup.R ran without error. Program code/generate_synthetic_data.R ran without error, but generated empty .rds files for the per-month/year YipitData listings. Program code/process_acs/process_acs.R ran without error. Program code/process_acs/nearby_demos_acs.R ran without error. Program code/generate_geo/generate_geo.R ran without error. Program code/generate_geo/load_zcta_data.R ran without error. Program code/process_infogroup/process_infogroup.R ran without error. Program code/process_yipit/add_brands.R broke down with error: empty .rds files caused a 0-row dataframe error. Program code/process_yipit/add_infogroup.R was not run. Program code/process_yipit/add_store_locations_w_infogroup.R was not run. Program code/process_yipit/separate_data.R was not run. Program code/process_yipit/compute_subset.R was not run. Program code/process_yipit/determine_top_chains_and_cuisines.R was not run. Program code/process_yipit/produce_DiD_data_v2.R was not run. Program code/process_numerator/collapse_static.R was not run. Program code/process_numerator/identify_static_trans.py was not run. Program code/process_numerator/identify_static_trans-summary.py was not run. Program code/process_numerator/combine_fact_summary.R was not run. Program code/process_numerator/combine_item_table.R broke down with error: data/numerator/2022-03-04/standard_nmr_feed_item_table.csv does not exist. Program code/process_numerator/add_category.R was not run. Program code/process_numerator/static_event_dataset.R was not run. Program code/process_numerator/produce_nonstatic_table.R was not run. Program code/process_numerator/prepare_regression_data.R was not run. Program code/describe_fee_caps/monthly_fee_cap_data.R was not run. Program code/describe_fee_caps/evolution_of_fee_caps_v2.R was not run. Program code/restaurant_choice_demos/nearby_demographics.R was not run. Program code/prepare_est_data/zcta_mapping.R was not run. Program code/prepare_est_data/zcta_resto_data.R was not run. Program code/prepare_est_data/produce_month_zip_level.R was not run. Program code/prepare_est_data/combine_data.R was not run. Program code/prepare_est_data/check_contradiction.R was not run. Program code/prepare_est_data/combine_months.R was not run. Program code/compute_pop_distribution/pop_dist_take2.R was not run eventually. In the first run this script broke down as it asked for a file in 'est_data'. Program code/zip_code_weights/construct_zip_code_weights.R was not run. Program code/numerator_menu_pricing/prepare_menu_data.R was not run. Program code/numerator_menu_pricing/prepare_geo_data.R was not run. Program code/numerator_menu_pricing/prepare_item_data.R was not run. Program code/numerator_menu_pricing/merge_item_level_data.R was not run. Program code/numerator_menu_pricing/convert_to_dta.do (Stata) was not run. Program code/numerator_menu_pricing/estimate_price_effects_disagg.do (Stata) was not run. Program code/numerator_menu_pricing/generate_item_level_reg_table.R was not run. Program code/numerator_menu_pricing/compute_price_indices.R was not run. All scripts in Stage 2–8 were not run.

9.3 Tables and Figures

Did not generate any output, therefore not able to verify anything. ## In-Text Numbers

- ☒ In-text numbers not verified.

- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code

9.4 Other Issues

- The README is very detailed and the codebase appears to be well-structured. However, the synthetic data option proved insufficient to run the full pipeline, as multiple scripts rely on proprietary data files that are either not generated or incorrectly generated by generate_synthetic_data.R.

10 Classification

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☐ partial reproduction (see above)
- ☒ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☐ Discrepancy in output (either figures or numbers in tables or text differ)
- ☒ Bugs in code that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ Code missing, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ Data preparation code missing should be checked if the code missing seems to be data preparation code
- ☒ Code not functional is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ Software not available to replicator may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ Insufficient time available to replicator is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ Data missing is marked when data should be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☒ Data not available is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ Missing README is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements

10.2 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this – please check!

Summary:

- Data files with PII indicators: 60000
- Variables flagged in data: 346913
- Code files with PII references: 212
- PII references in code: 5174

10.2.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	ACSDT5Y2019.B01003_data_with_overlays_2021-06-12T231340.csv	1	name
Data	ACSST5Y2019.S0101_data_with_overlays_2021-11-06T184610.csv	1	name

File Type	File	Variables/References	PII Categories
Data	ACSST5Y2019.S1201_data_with_overlays_2021-11-14T184552.csv	1	name
Data	ACSST5Y2019.S1501_data_with_overlays_2021-11-08T185630.csv	1	name
Data	ACSST5Y2019.S1901_data_with_overlays_2021-11-04T111610.csv	1	name
Data	ACS_one_year_2021_v3.csv	1	child
Data	MC_results.csv	1	lat
Data	0xCGRT_US_latest.csv	5	country, name, school, lat
Data	ZCTAs.csv	2	lat, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon
Data	all_data.csv	7	name, lat, loc, lon

See [Appendix](#) for detailed listing of all flagged instances.